Learning Over-Parametrized Two-Layer ReLU Networks beyond NTK

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COLT 2020

Paper 63 | Ref 91 | Tags: D, A / F, B | Must read

The paper in one minute

Question: Can trained neural networks learn a target that polynomial-size fixed-feature or NTK methods provably cannot learn as accurately?
Model/Method: A two-layer teacher with orthonormal directions, absolute-value activation and Gaussian inputs; a wide student trained by a *two-stage truncated gradient descent* method.
Finding: The student attains population error $O(d^{-1-Q})$ for a small constant $Q > 0$ with polynomial resources, whereas the specified polynomial-size kernel/feature methods have $\Omega(1/d)$ error lower bounds.
Why we care: Feature evolution can produce provable benefits unavailable to the NTK's fixed-kernel approximation.

1. What exactly is the NTK limitation?

A kernel method uses a fixed feature map $\phi(x)$ and learns only linear coefficients c, giving $f(x) = c^\top \phi(x)$. In the neural tangent kernel (NTK) approximation the effective features are the network's derivatives at initialization; parameters can move, but the learned predictor is modeled by a *fixed* kernel. A nonlinear neural network can instead modify the feature directions themselves. Li, Ma and Zhang construct a teacher–student problem where this extra freedom matters quantitatively.

The special target used in the theorem

The input $x \sim \mathcal{N}(0, I_d)$ and the teacher is

$$f^*(x) = \sum_{j=1}^{d} a_j |w_j^{*\top} x|, \qquad \{w_j^*\}_{j=1}^d \text{ orthonormal}, \quad a_j \geq 0.$$

Because $|z| = \mathrm{ReLU}(z) + \mathrm{ReLU}(-z)$, this absolute-value teacher can be represented by paired ReLU units. The nonnegative coefficients are bounded in the paper within a condition-number parameter κ.

Worked example: what low-order moments can miss

Take $d = 2$, teacher directions e_1, e_2, and $a_1 = a_2 = 1/2$. Then

$$f^*(x) = \tfrac{1}{2}(|x_1| + |x_2|).$$

For unit directions e_1, e_2, the second-moment matrix $\frac{1}{2}(e_1 e_1^\top + e_2 e_2^\top) = I/2$ treats every rotation similarly; it does not identify the individual coordinate axes. A fourth-order tensor contains $\frac{1}{2}(e_1^{\otimes 4} + e_2^{\otimes 4})$, which can reveal the two axes under suitable uniqueness conditions. This simplified calculation illustrates why recovering *higher-order structure* can require more than kernel-style linearized learning.

Big idea: An overparameterized network can reorganize the distribution of feature directions; a fixed NTK approximation cannot perform the same feature learning.

2. How does the proof establish a separation?

Algorithm 1 is not unrestricted vanilla GD

The theoretical procedure is *truncated gradient descent*: neurons with norm above a stage-dependent threshold are not updated. It uses small random Gaussian initial weights, a wide ReLU student with norm-dependent output coefficients, and two stages with different truncation levels. The truncation is technically important; do not replace the guaranteed method with arbitrary SGD or ordinary GD.

Stage-like feature discovery

First, training matches lower-order population statistics (the zeroth- and second-order tensor components). Then higher-order tensors, particularly the fourth-order structure, resolve the teacher's orthogonal directions. The authors' analysis features an initial slow plateau followed by faster progress, consistent with their numerical illustrations.

The two statements that must be compared

The learning theorem yields, under Gaussian input, orthonormal target weights, polynomially many neurons and samples, and specified truncation/step choices,

$$L(\widehat{W}) \le O(d^{-1-Q}), \qquad 0 < Q < 1/100.$$

Theorem 2 proves that a polynomial-size fixed feature map, or a kernel method given polynomial samples, incurs a lower bound of the form

$$L_{\text{kernel}} \ge \Omega(1/d)$$

for a suitable family of target weights. Hence the neural network achieves an asymptotically smaller *population* loss in this construction.

Worked comparison for intuition

With $Q = 0.05$ (illustrative), $d^{-1-Q}/d^{-1} = d^{-0.05}$. At $d = 10^4$ this ratio is $10^{-0.2} \approx 0.63$: the asymptotic ratio shrinks with d. Big-O and Ω contain constants, so this is *not* a predicted experiment or numerical guarantee.

> **Interpret theorem scope exactly:** The separation is for a special Gaussian teacher class and resource regime; it is not proof that every trainable neural network outperforms every kernel method on realistic datasets.

3. Contributions, assumptions, and limitations

Contributions. Provable population-risk improvement beyond polynomial-resource fixed-kernel models, a stage-by-stage explanation via higher-order tensor decomposition, and a finite-sample/finite-width transfer from a population dynamical analysis. **Assumptions.** Gaussian inputs; teacher weights forming an orthonormal basis; controlled positive teacher coefficients; particular two-layer ReLU student parameterization; large polynomial width and sample counts; carefully chosen learning rates and truncation thresholds. **Limitations.** Truncated GD differs from unrestricted GD in practice. The comparison is asymptotic and target-specific; the target/teacher is highly structured. The results do not prove global training convergence for arbitrary deep ReLU networks, nor superiority over every adaptive kernel technique.

Cross-paper connections

Mei et al. (Ref 89) describes moving neuron distributions, with kernel regression as a limiting regime. **Arora et al. (Ref 87)** explains optimization in a nearly frozen NTK regime; Li et al. prove why that regime can be insufficient. **Saxe (2014)** provides another stage-like mode-learning explanation in deep linear networks, though the mathematical models differ. **Xu–Du (Ref 92)** cautions that feature freedom and extra neurons do not necessarily speed convergence.

Quick reference

Object	Interpretation
$f^*(x) = \sum_j a_j \lvert {w_j^*}^\top x \rvert$	Structured Gaussian teacher with orthogonal directions.
Truncated GD	Gradient update gated by neuron-norm thresholds.
Zeroth/second moments	Initial coarse target information.
Fourth-order tensor	Helps identify orthogonal teacher directions.
$O(d^{-1-Q})$ vs $\Omega(1/d)$	Special neural-network vs polynomial-resource kernel loss comparison.

Papers Master note

Li–Ma–Zhang (2020) show that a wide two-layer ReLU student trained by staged truncated GD learns a special Gaussian orthogonal teacher to population error $O(d^{-1-Q})$, while polynomial-size kernel/fixed-feature methods face $\Omega(1/d)$ lower bounds. The separation illustrates feature learning beyond a fixed NTK regime.

Source: COLT 2020 original, Algorithm 1, Theorems 1–2, Section 2. Worked moment example pedagogical.